L'Equation de Black-Scholes : Dérivation et résolution

I- Dérivation de l'équation

* Hypothèses

- On est en temps continu
- Les taux d'intérêts sont constants
- Pas de dividendes
- Pas de coût de transaction
- Les actifs sont parfaitement divisibles
- La vente à découvert est autorisée
- Il n'y a pas d'opportunité d'arbitrage

Le processus stochastique de l'actif sous-jacent S_t suit un mouvement Brownien géométrique

$$dS_t = S_t(\mu dt + \sigma dB_t) \quad ; \quad \mu \text{ et } \sigma \text{ sont constants}$$

μ : rendement attendu
σ = volatilité de l'actif.
B_t : Mouvement Brownien

Soit Π la valeur d'un portefeuille dans lequel on a une position long sur option de valeur $V(t, S_t)$ et une position short sur Δ unité de notre actif sous-jacent.

$$\Pi(t,S_t) = V(t,S_t) - \Delta S_t$$

La variation de la valeur du portefeuille entre t et t+dt est:

$$d\Pi = dV(S_t,t) - \Delta\, dS_t$$

V étant une fonction stochastique de S_t, on peut appliquer le lemme d'Itô en admettant que $V \in \mathcal{C}^{1\times 2}$.

$$dV(t,S_t) = \frac{\partial}{\partial t}V(t,S_t)\,dt + \frac{\partial}{\partial S}V(t,S_t)\,dS_t + \frac{1}{2}\frac{\partial^2}{\partial S^2}V(t,S_t)\,d\langle S,S\rangle_t$$

$$dV(t,S_t) = \frac{\partial}{\partial t}V(t,S_t)\,dt + \frac{\partial V}{\partial S}(t,S_t)\,dS_t + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2}{\partial S^2}V(t,S_t)\,dt$$

Donc

$$dV = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S^2}\right)dt + \frac{\partial V}{\partial S}\,dS_t$$

Il s'en suit que

$$d\Pi = \underbrace{\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S^2}\right)dt}_{\text{déterministe}} + \underbrace{\left(\frac{\partial V}{\partial S} - \Delta\right)dS_t}_{\text{aléatoire}}$$

La partie aléatoire represente le risque dans notre portefeuille et on l'élimine en prenant $\Delta = \frac{\partial V}{\partial S}$ (Delta).

En prenant $\Delta = \frac{\partial V}{\partial S}$, $d\Pi$ devient

$$d\Pi(t,S_t) = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S^2}\right)dt$$

Notre portefeuille est désormais sans risque.

Par le principe de non-arbitrage, on sait que $d\Pi$ doit être au taux qu'on pourrait obtenir si on met le même montant dans un compte sans risque, donc

$$d\Pi = r\Pi\,dt$$

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} = r(V - \Delta S_t)$$

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} = rV - r\frac{\partial V}{\partial S} S_t$$

On a donc que

$$\boxed{rV = \frac{\partial V}{\partial t} + rS_t\frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2}}$$

(Equation de Black Scholes)

Et si notre actif payait des dividendes ?

$$rV = \frac{\partial V}{\partial t} + (r-D)\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2}$$

II- Résolution de B-S (Cas d'un call Européen)

Supposons que $f(x,t)$ est solution de l'équation

$$\begin{cases} rf = \frac{\partial}{\partial t}f + rx\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2 x^2 \frac{\partial^2}{\partial x^2}f \\ f(T,x) = h(x) = (x-K)^+ \end{cases}$$

(Condition terminale).

Alors la fonction $g(t,y)$ définie par

$$g(t,y) = e^{rt} f\left(T-t, e^{\sigma y + (\frac{\sigma^2}{2} - r)t}\right)$$

résout l'équation de la chaleur avec condition initiale

$\psi(y) = h(e^{\sigma y})$ c-à-d

$$\begin{cases} \dfrac{\partial g}{\partial t}(t,y) = \dfrac{1}{2}\dfrac{\partial^2}{\partial y^2} g(t,y) \\ \\ g(0,y) = h(e^{\sigma y}) \end{cases}$$

La solution de l'équation de la chaleur nous permettra de résoudre l'EDP de B-S.

* Equation de la chaleur

$$\frac{\partial \varphi}{\partial t} = \frac{1}{2}\frac{\partial^2 \varphi}{\partial y}$$

La solution fondamentale de cette équation est la densité d'une gaussienne donnée par

$$\varphi(t,y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{1}{2t}y^2}, \quad y \in \mathbb{R}.$$

$$\frac{\partial \varphi}{\partial t} = \frac{\partial}{\partial t}\left(\frac{1}{(2\pi t)^{1/2}}\right) e^{-\frac{1}{2t}y^2} + \left(\frac{1}{\sqrt{2\pi t}}\right)\frac{\partial}{\partial t}\left(\frac{-y^2}{2t}\right) e^{-\frac{1}{2t}y^2}$$

$$= \left(\frac{-\frac{\cancel{2}\pi}{\cancel{2}\sqrt{2\pi t}}}{2\pi t}\right) e^{-\frac{1}{2t}y^2} + \left(\frac{1}{\sqrt{2\pi t}}\right)\left(-\frac{y^2}{2}\left(-\frac{1}{t^2}\right)\right) e^{-\frac{1}{2t}y^2}$$

$$= \left(\frac{-\cancel{\pi}}{\sqrt{2\pi t}} \times \frac{1}{2\cancel{\pi} t}\right) e^{-\frac{1}{2t}y^2} + \left(\frac{1}{2\pi t}\right)\left(\frac{y^2}{2t^2}\right) e^{-\frac{1}{2t}y^2}$$

$$= \left(-\frac{1}{2t} + \frac{y^2}{2t^2}\right) \frac{1}{\sqrt{2\pi t}} e^{-\frac{1}{2t}y^2}$$

$$\frac{\partial}{\partial t}\varphi(t,y) = \left(-\frac{1}{2t} + \frac{y^2}{2t^2}\right)\varphi(t,y).$$

$$\frac{\partial^2}{\partial y^2}\varphi(t,y) = \frac{\partial}{\partial y}\left(\frac{1}{\sqrt{2\pi t}}\left(-\frac{y}{t}\right).\ e^{-y^2/2t}\right)$$

$$= \frac{1}{\sqrt{2\pi t}}\left(-\frac{1}{t}e^{-y^2/2t} + \left(-\frac{y}{t}\right).\left(-\frac{y}{t}\right)e^{-\frac{y^2}{2t}}\right)$$

$$= \left(-\frac{1}{t} + \frac{y^2}{t}\right)\frac{1}{\sqrt{2\pi t}}\ e^{-\frac{y^2}{2t}}$$

Donc

$$\frac{1}{2}\frac{\partial^2}{\partial y^2}\varphi(t,y) = \left(-\frac{1}{2t} + \frac{y^2}{t}\right)\varphi(t,y). \quad ; t>0,\ y\in\mathbb{R}.$$

$\varphi(t,y)$ est donc bien solution.

L'équation
$$\begin{cases} \frac{\partial}{\partial t}g(t,y) = \frac{1}{2}\frac{\partial^2}{\partial y^2}g(t,y) \\ g(0,y) = \psi(y) \quad \text{(Condition initiale continue)} \end{cases}$$

a pour solution $g(t,y) = \int_{\mathbb{R}} \psi(z).\ e^{-(y-z)^2/2t}\frac{dz}{\sqrt{2\pi t}}$, $y\in\mathbb{R}$, $t>0$

$$g(t,y) = \int_{\mathbb{R}} \psi(z)\ e^{-(y-z)^2/2t}\ \frac{dz}{\sqrt{2\pi t}} \quad ;$$

On pose $x = z-y$ et ~~on~~ obtient

$$g(t,y) = \int_{\mathbb{R}} \psi(x+y)\ e^{-\frac{x^2}{2t}}\ \frac{dx}{\sqrt{2\pi t}}$$

$$= \mathbb{E}(\psi(y+B_t)) \quad ; B_t \sim \mathcal{N}(0,t)$$

$$g(t,y) = \mathbb{E}(\psi(y-B_t)) \quad B_t = -B_t \text{ (loi)}.$$

En utilisant le lemme d'Itô, on a

$$g(t,y) = \mathbb{E}(\varphi(y-B_t))$$
$$= \mathbb{E}\Big(\varphi(y-\underbrace{B_0}_{=0}) + \int_0^t \varphi'(y-B_s)\,dB_s + \frac{1}{2}\int_0^t \varphi''(y-B_s)\,ds\Big)$$
$$= \mathbb{E}(\varphi(y)) + \underbrace{\mathbb{E}\Big(\int_0^t \varphi'(y-B_s)\,dB_s\Big)}_{=0} + \frac{1}{2}\,\mathbb{E}\Big(\int_0^t \varphi''(y-B_s)\,ds\Big)$$
$= 0$ (Espérance d'une intégrale d'Itô)

$$= \varphi(y) + \frac{1}{2}\,\mathbb{E}\Big(\int_0^t \varphi''(y-B_s)\,ds\Big)$$

$$g(t,y) = \varphi(y) + \frac{1}{2}\int_0^t \mathbb{E}(\varphi''(y-B_s))\,ds.$$

or $$\frac{\partial}{\partial y} g(t,y) = \frac{\partial}{\partial y}\big(\mathbb{E}(\varphi(y-B_t))\big)$$
$$\frac{\partial}{\partial y} g = \mathbb{E}(\varphi'(y-B_t)) \quad \text{et} \quad \frac{\partial^2}{\partial y^2} g = \mathbb{E}(\varphi''(y-B_t))$$

Donc $$\underline{g(t,y) = \varphi(y) + \frac{1}{2}\int_0^t \frac{\partial^2}{\partial y^2}\mathbb{E}(\varphi(y-B_s))\,ds}$$

$$\frac{\partial g(t,y)}{\partial t} = \frac{1}{2}\frac{\partial}{\partial t}\Big(\int_0^t \frac{\partial^2}{\partial y^2}\mathbb{E}(\varphi(y-B_s))\,ds$$
$$\frac{\partial g}{\partial t} = \frac{1}{2}\frac{\partial^2}{\partial y^2}\mathbb{E}(\varphi(y-B_t))$$
$$\underline{\underline{\frac{\partial g}{\partial t} = \frac{1}{2}\frac{\partial^2}{\partial y^2} g}}$$

$$g(0,y) = \mathbb{E}(\varphi(y-B_0)) = \mathbb{E}(\varphi(y)) = \varphi(y).$$

Vérifions maintenant que $g(t,y) = e^{rt} f(T-t, e^{\sigma y + (\frac{\sigma^2}{2} - r)t})$ vérifie l'équation de la chaleur.

Posons $s = T-t$ et $x = e^{\sigma y + (\frac{\sigma^2}{2} - r)t}$

$$\frac{\partial s}{\partial t} = -1 \qquad \frac{\partial x}{\partial t} = (\frac{\sigma^2}{2} - r) \cdot x \quad ; \quad \frac{\partial x}{\partial y} = \sigma x.$$

$$\frac{\partial g}{\partial t} = r e^{rt} \cdot f + e^{rt} \cdot \left(-\frac{\partial}{\partial s} f + \left(\frac{\sigma^2}{2} - r \right) x \frac{\partial}{\partial x} f \right)$$

$$= e^{rt} \left(rf - \frac{\partial}{\partial s} f + \frac{1}{2} \sigma^2 x \frac{\partial f}{\partial x} - rx \frac{\partial f}{\partial x} \right)$$

$$\frac{\partial g}{\partial t} = e^{rt} \left(\frac{1}{2} \sigma^2 x^2 \frac{\partial^2}{\partial x^2} f + \frac{1}{2} \sigma^2 x \frac{\partial}{\partial x} f \right)$$

$$\frac{\partial g}{\partial y} = e^{rt} \frac{\partial}{\partial y} f = e^{rt} \frac{\partial f}{\partial x} \cdot \frac{\partial y}{\partial x} = \sigma e^{rt} x \frac{\partial f}{\partial x}$$

$$\frac{\partial^2 g}{\partial y^2} = \frac{\partial}{\partial y} \left(\sigma e^{rt} x \frac{\partial f}{\partial x} \right)$$

$$= \sigma e^{rt} \frac{\partial}{\partial y} \left(x \cdot \frac{\partial f}{\partial x} \right)$$

$$= \sigma e^{rt} \left(\sigma x \cdot \frac{\partial f}{\partial x} + x \frac{\partial x}{\partial y} \cdot \frac{\partial^2 f}{\partial x^2} \right)$$

$$\frac{\partial^2 g}{\partial y^2} = \sigma e^{rt} \left(\sigma x \frac{\partial f}{\partial x} + \sigma x^2 \frac{\partial^2 f}{\partial x^2} \right)$$

$$\frac{1}{2} \frac{\partial^2 g}{\partial y^2} = e^{rt} \left(\frac{1}{2} \sigma^2 x \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 f}{\partial x^2} \right) = \frac{\partial}{\partial t} g.$$

g est bien solution

$$g(0,y) = e^{r \times 0} f(T-0, e^{\sigma y}) = f(T, e^{\sigma y}) = h(e^{\sigma y}).$$

Passons par la résolution de l'EDP de B-S par inversion de g.

$$S = T - t \quad , \quad x = e^{\sigma y + (\frac{\sigma^2}{2} - r)t}$$

$$f(s,x) = e^{-(T-s)r} g\left(T-s, \frac{1}{\sigma}\left(\log(x) - (\frac{\sigma^2}{2} - r)(T-s)\right)\right)$$

et $h(x) = \psi\left(\frac{1}{\sigma}\log(x)\right)$, $x>0$ ou $\psi(y) = h(e^{\sigma y})$

En utilisant la solution $\int_{\mathbb{R}} \psi(z) e^{-(y-z)^2/2t} \frac{dz}{\sqrt{2\pi t}}$ et le fait que $\psi(y) = h(e^{y})$, on a:

$$f(t,x) = e^{-(T-t)r} \int_{\mathbb{R}} \psi\left[\frac{1}{\sigma}\left(\log(x) - (\frac{\sigma^2}{2} - r)(T-t)\right) + z\right] e^{-\frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}}$$

$$= e^{-(T-t)r} \int_{\mathbb{R}} h\left(e^{\sigma z + \log(x) - (\frac{\sigma^2}{2} - r)(T-t)}\right) e^{-\frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}}$$

$$= e^{-r(T-t)} \int_{\mathbb{R}} h\left(x e^{\sigma z - (\frac{\sigma^2}{2} - r)(T-t)}\right) e^{-\frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}} \quad ; \quad e^{\log(x)} = x$$

$$h(x) = (x - K)^+$$

$$f(t,x) = e^{-r(T-t)} \int_{\mathbb{R}} \left(x e^{\sigma z - (\frac{\sigma^2}{2} - r)(T-t)} - K\right)^+ e^{-\frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}}$$

$$e^{\sigma z - (\frac{\sigma^2}{2} - r)(T-t)} > \frac{K}{x}$$

$$\sigma z - (\frac{\sigma^2}{2} - r)(T-t) > \log(\frac{K}{x})$$

$$\sigma z > \log(\frac{K}{x}) + (\frac{\sigma^2}{2} - r)(T-t)$$

$$z > \underbrace{\frac{1}{\sigma}\left[\log(\frac{K}{x}) + (\frac{\sigma^2}{2} - r)(T-t)\right]}_{a.}$$

$$f(t,x) = e^{-r(T-t)} \int_a^{+\infty} x\, e^{\sigma z - (\frac{\sigma^2}{2} - r)(T-t)} \cdot e^{-z^2/2(T-t)} \frac{dz}{\sqrt{2(T-t)\pi}}$$

$$- K e^{-r(T-t)} \int_a^{+\infty} e^{-\frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}}$$

Posons $\quad I_1 = x e^{-r(T-t)} \int_a^{+\infty} e^{\sigma z - (\frac{\sigma^2}{2} - r)(T-t) - \frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}}$

$$I_2 = K e^{-r(T-t)} \int_a^{+\infty} e^{-\frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}}$$

$$I_1 = x e^{-r(T-t)} \int_a^{+\infty} e^{\frac{-1}{2(T-t)}\left(z^2 - 2(T-t)\sigma z + (T-t)^2\sigma^2\right)} \cdot e^{r(T-t)} \frac{dz}{\sqrt{2(T-t)\pi}}$$

$$= x \int_a^{+\infty} e^{-\frac{1}{2(T-t)}(z - \sigma(T-t))^2} \frac{dz}{\sqrt{2(T-t)\pi}}$$

Posons $\quad u = z - \sigma(T-t)$

$$a - \sigma(T-t) = \frac{1}{\sigma}\left(\log\left(\frac{K}{x}\right) + \left(\frac{\sigma^2}{2} - r\right)(T-t)\right) - \sigma(T-t)$$

$$= \frac{1}{\sigma}\left(\log\left(\frac{K}{x}\right) + \left(\frac{\sigma^2}{2} - \sigma^2 - r\right)(T-t)\right)$$

$$\underline{b = \frac{1}{\sigma}\left(\log\left(\frac{K}{x}\right) + \left(-\frac{\sigma^2}{2} - r\right)(T-t)\right)} = -\frac{1}{\sigma}\left(\log\left(\frac{x}{K}\right) + \left(\frac{\sigma^2}{2} + r\right)(T-t)\right)$$

$$I_1 = x \int_b^{+\infty} e^{-\frac{1}{2(T-t)}u^2} \frac{du}{\sqrt{2(T-t)\pi}}$$

Posons $\quad v = \frac{u}{\sqrt{T-t}} \implies dv = \frac{1}{\sqrt{T-t}}\, du$

$$c = -\frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{x}{K}\right) + \left(\frac{\sigma^2}{2} + r\right)(T-t)\right) = -d_+$$

$$I_2 = x\int_c^{+\infty} \frac{e^{-\frac{1}{2}v^2}}{\sqrt{2\pi}}\,dv$$

$$I_1 = x\int_{-d_+}^{+\infty} \frac{e^{-\frac{1}{2}v^2}}{\sqrt{2\pi}}\,dv$$

$$= x\left(1 - \Phi(-d_+)\right)$$

$$= x\,\Phi(-(-d_+))$$

$$\underline{\underline{I_1 = x\,\Phi(d_+)}}$$

$$I_2 = K e^{-r(T-t)} \int_a^{+\infty} e^{-\frac{z^2}{2(T-t)}} \frac{dz}{\sqrt{2(T-t)\pi}}$$

$$a = \frac{1}{\sigma}\left(\log\left(\frac{K}{x}\right) + \left(\frac{\sigma^2}{2} - r\right)(T-t)\right)$$

$$= -\frac{1}{\sigma}\left(\log\left(\frac{x}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)(T-t)\right)$$

$$u = \frac{z}{\sqrt{T-t}} \Rightarrow du = \frac{dz}{\sqrt{T-t}}$$

$$I_2 = K e^{-r(T-t)} \int_b^{\infty} e^{-\frac{1}{2}u^2} \frac{du}{\sqrt{2\pi}}$$

$$= K e^{-r(T-t)} \left(1 - \Phi(-d_-)\right) \qquad d_- = \frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{x}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)(T-t)\right)$$

$$= K e^{-r(T-t)} \left(\Phi(-(-d_-))\right)$$

$$\underline{I_2 = K e^{-r(T-t)}\,\Phi(d_-)}$$

$$\underline{f(x,t) = x\,\Phi(d_+) - K e^{-r(T-t)}\,\Phi(d_-)}$$

III - Les grecs

$$C = S\,\Phi(d_+) - K e^{-r(T-t)}\,\Phi(d_-)$$

$$d_+ = \frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)\right)$$

$$d_- = d_+ - \sigma\sqrt{T-t}$$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

$$\frac{f(d_-)}{f(d_+)} = \frac{e^{-\frac{d_-^2}{2}}}{e^{-\frac{d_+^2}{2}}} = e^{\left(\frac{d_+^2}{2} - \frac{d_-^2}{2}\right)}$$

$$\begin{aligned} d_+^2 - d_-^2 &= d_+^2 - (d_+ - \sigma\sqrt{T-t})^2 \\ &= \cancel{d_+^2} - (\cancel{d_+^2} - 2(\sigma\sqrt{T-t})\,d_+ + \sigma^2(T-t)) \\ &= 2 d_+ \sigma\sqrt{T-t} - \sigma^2(T-t) \\ &= 2\left(\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)\right) - \sigma^2(T-t) \end{aligned}$$

$$d_+^2 - d_-^2 = 2\ln\left(\frac{S}{K}\right) + 2r(T-t)$$

$$\frac{f(d_-)}{f(d_+)} = e^{\ln\left(\frac{S}{K}\right) + 2r(T-t)} = \frac{S}{K} e^{2r(T-t)}$$

$$\underline{S f(d_+) = K e^{-2r(T-t)} f(d_-)}$$

* **Delta** (sensibilité au spot)

$$\Delta = \frac{\partial C}{\partial S}$$

$$\frac{\partial d_\pm}{\partial S} = \frac{\partial}{\partial S}\left(\frac{1}{\sigma\sqrt{T-t}} \left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2} + r\right)(T-t) \right)\right)$$

$$= \frac{1}{\sigma\sqrt{T-t}} \cdot \frac{\frac{1}{K}}{\frac{S}{K}} = \frac{1}{\sigma\sqrt{T-t}} \frac{1}{\cancel{K}} \times \frac{\cancel{K}}{S} = \frac{1}{\sigma S\sqrt{T-t}}$$

$$\frac{\partial d_+}{\partial S} = \frac{\partial}{\partial S} d_-$$

$$\Delta = \frac{\partial}{\partial S}\left(S\,\Phi(d_+) - K e^{-r(T-t)} \Phi(d_-) \right)$$

$$= \Phi(d_+) + \cancel{S} \times \frac{1}{\sigma \cancel{S}\sqrt{T-t}} \Phi'(d_+) - K e^{-r(T-t)} \times \frac{1}{\sigma S\sqrt{T-t}} \Phi'(d_-)$$

$$= \Phi(d_+) + \frac{\Phi'(d_+)}{\sigma\sqrt{T-t}} - \frac{\cancel{S}\,\Phi'(d_+)}{\sigma \cancel{S}\sqrt{T-t}}$$

$$\underline{\underline{\Delta = \Phi(d_+)}}$$

* **Gamma** (Courbure)

$$\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\partial}{\partial S}\left(\Phi(d_+)\right)$$

$$= \frac{1}{\sigma S\sqrt{T-t}} \phi'(d_+)$$

* **Vega** (sensibilité à la vol)

$$\text{Vega} = \frac{\partial C}{\partial \sigma}$$

$$\text{Vega} = \frac{\partial}{\partial \sigma}\left(S\,\Phi(d_+) - K e^{-r(T-t)}\,\Phi(d_-)\right)$$

$$\frac{\partial}{\partial \sigma} d_+ = \frac{\partial}{\partial \sigma}\left(\frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)\right)\right)$$

$$= -\frac{1}{\sigma^2\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)\right) + \frac{1}{\cancel{\sigma}\sqrt{T-t}}\left(\cancel{\sigma}(T-t)\right)$$

$$\frac{\partial d_+}{\partial \sigma} = \sqrt{T-t} - \frac{1}{\sigma^2\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)\right)$$

$$\frac{\partial d_-}{\partial \sigma} = \frac{\partial}{\partial \sigma}\left(d_+ - \sigma\sqrt{T-t}\right)$$

$$\frac{\partial d_-}{\partial \sigma} = \frac{\partial}{\partial \sigma} d_+ - \sqrt{T-t} = -\frac{1}{\sigma^2\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)\right)$$

$$\frac{\partial C}{\partial \sigma} = S\left(\sqrt{T-t} - \frac{1}{\sigma^2\sqrt{(T-t)}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)\right)\right)\Phi'(d_+) + \frac{K e^{-r(T-t)}}{\sigma^2\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}-r\right)(T-t)\right)\Phi'(d_-)$$

$$= S\left(\sqrt{T-t} - \left(\frac{\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)}{\sigma^2\sqrt{T-t}}\right)\right)\Phi'(d_+) + \frac{S\,\Phi'(d_+)}{\sigma^2\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)\right)$$

$$\frac{\partial C}{\partial \sigma} = S\sqrt{T-t}\;\Phi'(d_+)$$

* **Theta** (sensibilité au temps)

$$\Theta = \frac{\partial C}{\partial t} = \frac{\partial}{\partial t}\left(S\,\Phi(d_+) - K e^{-r(T-t)}\,\Phi(d_-)\right)$$

$$\frac{\partial d_+}{\partial t} = \frac{\partial}{\partial t}\left(\frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2}+r\right)(T-t)\right)\right)$$

$$\frac{\partial d_+}{\partial t} = \frac{1}{\sigma}\log\left(\frac{S}{K}\right) \times \left(\frac{-\frac{1}{2\sqrt{T-t}}}{T-t}\right) - \left(\frac{\sigma^2}{2} + r\right)$$

$$= \frac{1}{2\sqrt{T-t}} \times \frac{1}{T-t} - \left(\frac{\sigma^2}{2} + r\right)$$

$$\frac{\partial d_+}{\partial t} = \frac{1}{2(T-t)\sqrt{T-t}} - \frac{\sigma^2}{2} - r$$

$$\frac{\partial d_-}{\partial t} = \frac{\partial}{\partial t}\left(d_+ - \sigma\sqrt{T-t}\right)$$

$$\frac{\partial d_-}{\partial t} = \frac{\partial d_+}{\partial t} + \frac{\sigma}{2\sqrt{T-t}} = \frac{1}{2\sqrt{T-t}}\left(\frac{1}{(T-t)} + \sigma\right) - \frac{\sigma^2}{2} - r$$

$$\frac{\partial C}{\partial t} = S\left(\frac{1}{2(T-t)\sqrt{T-t}} - \frac{\sigma^2}{2} - r\right)\Phi'(d_+) - rKe^{-r(T-t)} \cdot \Phi(d_-)$$

$$- Ke^{-r(T-t)} \cdot \left(\frac{1}{2\sqrt{T-t}}\left(\frac{1}{T-t} + \sigma\right) - \frac{\sigma^2}{2} - r\right)\Phi'(d_-)$$

$$\frac{\partial C}{\partial t} = \left(\frac{1}{2(T-t)\sqrt{T-t}} - \frac{1}{2\sqrt{T-t}\,(T-t)} - \frac{\sigma}{2\sqrt{T-t}}\right) S\,\Phi'(d_+)$$

$$- Ke^{-r(T-t)}\,\Phi(d_-)$$

$$\frac{\partial C}{\partial t} = -\frac{\sigma}{2\sqrt{T-t}} S\,\Phi'(d_+) - rKe^{-r(T-t)}\,\Phi(d_-)$$

$$\theta = \frac{-\sigma S}{2\sqrt{T-t}}\Phi'(d_+) - rKe^{-r(T-t)}\,\Phi(d_-)$$

* Rho (Sensibilité au taux)

$$\rho = \frac{\partial C}{\partial r} = \frac{\partial}{\partial r}\left(S\,\Phi(d_+) - Ke^{-r(T-t)}\,\Phi(d_-)\right)$$

$$\frac{\partial d_+}{\partial r} = \frac{\partial}{\partial r}\left(\frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S}{K}\right) + \left(\frac{\sigma^2}{2} + r\right)(T-t)\right)\right) = (T-t)$$

$$\frac{\partial d_-}{\partial r} = \frac{\partial}{\partial r}\left(d_+ - \sigma\sqrt{T-t}\right) = \frac{\partial d_+}{\partial r}$$

$$\rho = \frac{\partial C}{\partial r} = \frac{\partial}{\partial r}\left(S\Phi(d_+) - Ke^{-r(T-t)}\Phi(d_-)\right)$$

$$= S(T-t)\Phi'(d_+) + K(T-t)e^{-r(T-t)}\Phi(d_-) - K(T-t)e^{-r(T-t)}\Phi'(d_-)$$

$$= S(T-t)\Phi'(d_+) - S(T-t)\Phi'(d_+) + K(T-t)e^{-r(T-t)}\Phi(d_-)$$

$$= K(T-t)e^{-r(T-t)}$$

$$\underline{\rho = \frac{\partial C}{\partial r} = K(T-t)e^{-r(T-t)}}\ \Phi(d_-)$$